

CHAPTER 1:

```

File Machine View Input Devices Help
1 2 3 4

PS> cys_714023149064@kali: /home/cys_714023149064

File Actions Edit View Help

Trash

top - 03:08:33 up 5 min, 2 users, load average: 0.28, 0.41, 0.24
Tasks: 166 total, 3 running, 163 sleeping, 0 stopped, 0 zombie
%Cpu(s): 17.2 us, 13.8 sy, 0.0 ni, 67.9 id, 0.4 wa, 0.0 hi, 0.7 si, 0.0 st
MiB Mem : 3922.3 total, 2803.0 free, 847.3 used, 489.8 buff/cache
MiB Swap: 1101.0 total, 1101.0 free, 0.0 used. 3075.0 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  704 root        20   0   418572  125824  61228 R   15.0   3.1   0:20.68 Xorg
  1126 cys_714+    20   0   418784  35868   22948 S   12.3   0.9   0:00.65 xfce4-notifyd
  1040 cys_714+    20   0   983208  125800  83692 R    3.3   3.1   0:04.12 xfwm4
  1081 cys_714+    20   0   460372  42964   33240 S    2.0   1.1   0:01.12 panel-1-whisker
   863 cys_714+    20   0   313832   9836    8812 S    0.7   0.2   0:00.11 gnome-keyring-d
top - 03:08:37 up 5 min, 2 users, load average: 0.26, 0.40, 0.24
Tasks: 166 total, 1 running, 165 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.7 us, 3.7 sy, 0.0 ni, 89.0 id, 0.2 wa, 0.0 hi, 3.5 si, 0.0 st
MiB Mem : 3922.3 total, 2795.7 free, 854.6 used, 489.9 buff/cache
MiB Swap: 1101.0 total, 1101.0 free, 0.0 used. 3067.8 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   704 root        20   0   430928  138104  61228 S    6.6   3.4   0:21.34 Xorg
    26 root        20   0         0         0      0 S    2.3   0.0   0:03.90 ksoftirqd/1
  1040 cys_714+    20   0   983208  125800  83692 S    2.0   3.1   0:04.26 xfwm4
   632 root        20   0   333608  18484   15796 S    1.7   0.5   0:02.59 NetworkManager
   970 cys_714+    20   0   216064   2996    2728 S    1.0   0.1   0:02.34 VBoxClient
  1491 cys_714+    20   0   465892  106064  91420 S    1.0   2.6   0:05.54 qterminal
  1088 cys_714+    20   0   286364   47044  19200 S    0.7   1.2   0:03.31 panel-13-cpugra
  1090 cys_714+    20   0   338096   28156  20964 S    0.7   0.7   0:01.77 panel-15-genmon
    40 root        20   0         0         0      0 S    0.3   0.0   0:00.06 kcompactd0
top - 03:09:07 up 6 min, 2 users, load average: 0.22, 0.38, 0.24
Tasks: 165 total, 1 running, 164 sleeping, 0 stopped, 0 zombie
%Cpu(s): 6.7 us, 7.3 sy, 0.0 ni, 84.7 id, 0.2 wa, 0.0 hi, 1.1 si, 0.0 st
MiB Mem : 3922.3 total, 2836.5 free, 813.8 used, 493.6 buff/cache
MiB Swap: 1101.0 total, 1101.0 free, 0.0 used. 3108.6 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   704 root        20   0   443864  125216  64792 S   10.9   3.1   0:25.11 Xorg
  1491 cys_714+    20   0   465932  106060  91416 S    4.6   2.6   0:06.03 qterminal
  1040 cys_714+    20   0   983208  125800  83692 S    2.6   3.1   0:04.92 xfwm4
  1088 cys_714+    20   0   286364   47044  19200 S    1.3   1.2   0:03.70 panel-13-cpugra
  1012 cys_714+    20   0   234180   7776    7008 S    0.7   0.2   0:00.61 at-spi2-registr
  1090 cys_714+    20   0   338096   28156  20964 S    0.7   0.7   0:01.91 panel-15-genmon
  1096 cys_714+    20   0   385764   39516  31336 S    0.7   1.0   0:00.51 panel-22-action
    26 root        20   0         0         0      0 S    0.3   0.0   0:03.98 ksoftirqd/1
    30 root        20   0         0         0      0 I    0.3   0.0   0:00.09 kworker/u10:0-flush-8:0
    47 root        20   0         0         0      0 I    0.3   0.0   0:00.98 kworker/1:1-events
   970 cys_714+    20   0   216064   2996    2728 S    0.3   0.1   0:02.58 VBoxClient
  1027 cys_714+    20   0   215652   3516    3120 S    0.3   0.1   0:00.33 VBoxClient
  1080 cys_714+    20   0   509704   60416  39396 S    0.3   1.5   0:05.61 xfdesktop
  1123 root        20   0   317616   9928    8136 S    0.3   0.2   0:00.61 upowerd

```

File Actions Edit View Help

```
(cys_714023149064@kali)~[~/linux-practice]
$ mkdir proj
```

```
(cys_714023149064@kali)~[~/linux-practice]
$ cd ~/linux-practice
```

```
(cys_714023149064@kali)~[~/linux-practice]
$ mkdir testfolder
```

```
(cys_714023149064@kali)~[~/linux-practice]
$ ls
```

backups backup.tar documents images music proj projects scripts testfolder videos

```
(cys_714023149064@kali)~[~/linux-practice]
$ ls -l
```

```
total 48
drwxrwxr-x 2 cys_714023149064 cys_714023149064 4096 May 14 12:32 backups
-rw-rw-r-- 1 cys_714023149064 cys_714023149064 10240 May 14 12:31 backup.tar
drwxrwxr-x 2 cys_714023149064 cys_714023149064 4096 May 14 12:30 documents
drwxrwxr-x 2 cys_714023149064 cys_714023149064 4096 May 14 12:32 images
drwxrwxr-x 2 cys_714023149064 cys_714023149064 4096 May 14 12:29 music
drwxrwxr-x 2 cys_714023149064 cys_714023149064 4096 May 14 12:34 proj
drwxrwxr-x 4 cys_714023149064 cys_714023149064 4096 May 14 12:31 projects
drwxrwxr-x 2 cys_714023149064 cys_714023149064 4096 May 14 12:28 scripts
drwxrwxr-x 2 cys_714023149064 cys_714023149064 4096 May 14 12:34 testfolder
drwxrwxr-x 2 cys_714023149064 cys_714023149064 4096 May 14 12:29 videos
```

```
(cys_714023149064@kali)~[~/linux-practice]
$ ls -a
```

. .. backups backup.tar documents images music proj projects scripts .secret

```
(cys_714023149064@kali)~[~/linux-practice]
$ touch sample.txt
```

```
(cys_714023149064@kali)~[~/linux-practice]
$ echo "Hello Linux" > sample.txt
```

```
(cys_714023149064@kali)~[~/linux-practice]
$ cat sample.txt
```

Hello Linux

```
(cys_714023149064@kali)~[~/linux-practice]
$ cp sample.txt copy.txt
```

```
(cys_714023149064@kali)~[~/linux-practice]
$ mv copy.txt renamed.txt
```

```
(cys_714023149064@kali)~[~/linux-practice]
$ ls
```

backups documents music projects sample.txt testfolder
backup.tar images proj renamed.txt scripts videos

CHAPTER 2

File Actions Edit View Help

```
(cys_714023149064@kali)~[/linux-practice]
```

```
$ mkdir proj
```

```
(cys_714023149064@kali)~[/linux-practice]
```

```
$ cd ~/linux-practice
```

```
(cys_714023149064@kali)~[/linux-practice]
```

```
$ mkdir testfolder
```

```
(cys_714023149064@kali)~[/linux-practice]
```

```
$ ls
```

```
backups  backup.tar  documents  images  music  proj  projects  scripts  testfolder  videos
```

```
(cys_714023149064@kali)~[/linux-practice]
```

```
$ ls -l
```

```
total 48
```

drwxrwxr-x	2	cys_714023149064	cys_714023149064	4096	May 14 12:32	backups
-rw-rw-r--	1	cys_714023149064	cys_714023149064	10240	May 14 12:31	backup.tar
drwxrwxr-x	2	cys_714023149064	cys_714023149064	4096	May 14 12:30	documents
drwxrwxr-x	2	cys_714023149064	cys_714023149064	4096	May 14 12:32	images
drwxrwxr-x	2	cys_714023149064	cys_714023149064	4096	May 14 12:29	music
drwxrwxr-x	2	cys_714023149064	cys_714023149064	4096	May 14 12:34	proj
drwxrwxr-x	4	cys_714023149064	cys_714023149064	4096	May 14 12:31	projects
drwxrwxr-x	2	cys_714023149064	cys_714023149064	4096	May 14 12:28	scripts
drwxrwxr-x	2	cys_714023149064	cys_714023149064	4096	May 14 12:34	testfolder
drwxrwxr-x	2	cys_714023149064	cys_714023149064	4096	May 14 12:29	videos

```
(cys_714023149064@kali)~[/linux-practice]
```

```
$ ls -la
```

```
.  ..  backups  backup.tar  documents  images  music  proj  projects  scripts  .secret
```

```
(cys_714023149064@kali)~[/linux-practice]
```

```
$ touch sample.txt
```

```
(cys_714023149064@kali)~[/linux-practice]
```

```
$ echo "Hello Linux" > sample.txt
```

```
(cys_714023149064@kali)~[/linux-practice]
```

```
$ cat sample.txt
```

```
Hello Linux
```

```
(cys_714023149064@kali)~[/linux-practice]
```

```
$ cp sample.txt copy.txt
```

```
(cys_714023149064@kali)~[/linux-practice]
```

```
$ mv copy.txt renamed.txt
```

```
(cys_714023149064@kali)~[/linux-practice]
```

```
$ ls
```

```
backups  documents  music  projects  sample.txt  testfolder  
backup.tar  images  proj  renamed.txt  scripts  videos
```

```
File Actions Edit View Help
(cys_714023149064@kali)-[~]
$ cd ~/linux-practice
(cys_714023149064@kali)-[~/linux-practice]
$ cat distros.txt
Linux
Kali
Ubuntu
Debian
Fedora
Arch

(cys_714023149064@kali)-[~/linux-practice]
$ more distros.txt
Linux
Kali
Ubuntu
Debian
Fedora
Arch

(cys_714023149064@kali)-[~/linux-practice]
$ tail distros.txt
Linux
Kali
Ubuntu
Debian
Fedora
Arch

(cys_714023149064@kali)-[~/linux-practice]
$ head distros.txt
Linux
Kali
Ubuntu
Debian
Fedora
Arch

(cys_714023149064@kali)-[~/linux-practice]
$ nano distros.txt

(cys_714023149064@kali)-[~/linux-practice]
$ vim distros.txt

(cys_714023149064@kali)-[~/linux-practice]
$ gedit distros.txt
Command 'gedit' not found, but can be installed with:
sudo apt install gedit
Do you want to install it? (N/y)N

(cys_714023149064@kali)-[~/linux-practice]
$ awk '{print $1}' distros.txt
Linux
```

CHAPTER 3:

```
Arch
(cys_714023149064@kali) - [~/linux-practice]
$ nano distros.txt

(cys_714023149064@kali) - [~/linux-practice]
$ vim distros.txt

(cys_714023149064@kali) - [~/linux-practice]
$ gedit distros.txt
Command 'gedit' not found, but can be installed with:
sudo apt install gedit
Do you want to install it? (N/y)N

(cys_714023149064@kali) - [~/linux-practice]
$ awk '{print $1}' distros.txt
Linux
Kali
Ubuntu
Debian
Fedora
Arch
hi
i

(cys_714023149064@kali) - [~/linux-practice]
$ sed 's/Linux/Parrot/' distros.txt
Parrot
Kali
Ubuntu
Debian
Fedora
Arch
hi
i

(cys_714023149064@kali) - [~/linux-practice]
$
```

File Actions Edit View Help

```
Linux
(cys_714023149064@kali)-[~/linux-practice]
$ ls -l permission.txt
-rw-rw-r-- 1 cys_714023149064 cys_714023149064 0 May 14 12:52 permission.txt
Fedora
(cys_714023149064@kali)-[~/linux-practice]
$ chmod 777 permission.txt

(cys_714023149064@kali)-[~/linux-practice]
$ ls -l permission.txt
-rwxrwxrwx 1 cys_714023149064 cys_714023149064 0 May 14 12:52 permission.txt

(cys_714023149064@kali)-[~/linux-practice]
$ sudo chown root:root permission.txt
[sudo] password for cys_714023149064:

(cys_714023149064@kali)-[~/linux-practice]
$ ls -l permission.txt
-rwxrwxrwx 1 root root 0 May 14 12:52 permission.txt

(cys_714023149064@kali)-[~/linux-practice]
$ sudo chgrp root permission.txt

(cys_714023149064@kali)-[~/linux-practice]
$ umask
002

(cys_714023149064@kali)-[~/linux-practice]
$ stat permission.txt
  File: permission.txt
  Size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 8,1      Inode: 1073141    Links: 1
Access: (0777/-rwxrwxrwx)  Uid: (    0/   root)   Gid: (    0/   root)
Access: 2026-05-14 12:52:31.400217226 -0400
Modify: 2026-05-14 12:52:31.400217226 -0400
Change: 2026-05-14 12:53:43.400199033 -0400
 Birth: 2026-05-14 12:52:31.400217226 -0400

(cys_714023149064@kali)-[~/linux-practice]
$ getfacl permission.txt
# file: permission.txt
# owner: root
# group: root
user::rwx
group::rwx
other::rwx

(cys_714023149064@kali)-[~/linux-practice]
$ setfacl -m u:${whoami}:r permission.txt
setfacl: permission.txt: Operation not permitted

(cys_714023149064@kali)-[~/linux-practice]
$
```

CHAPTER 4

File Actions Edit View Help

Linux

```
(cys_714023149064@kali)-[~/linux-practice]
$ sudo chattr +i permission.txt
```

Debian

```
(cys_714023149064@kali)-[~/linux-practice]
$ sudo usermod -L testuser
usermod: user 'testuser' does not exist
```

```
(cys_714023149064@kali)-[~/linux-practice]
$ sudo usermod -U testuser
usermod: user 'testuser' does not exist
```

```
(cys_714023149064@kali)-[~/linux-practice]
$ sudo useradd testuser
```

```
(cys_714023149064@kali)-[~/linux-practice]
$ sudo usermod -U testuser
usermod: unlocking the user's password would result in a passwordless account.
You should set a password with usermod -p to unlock this user's password.
```

```
(cys_714023149064@kali)-[~/linux-practice]
$
```

File Actions Edit View Help

Linux

(cys_714023149064@kali)-[~/linux-practice]

\$ ifconfig

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.36.179.18 netmask 255.255.255.0 broadcast 10.36.179.255
    inet6 fe80::a00:27ff:fe13:ae22 prefixlen 64 scopeid 0x20<link>
    inet6 2401:4900:67c3:36b5:a00:27ff:fe13:ae22 prefixlen 64 scopeid 0x0<global>
    inet6 2401:4900:67c3:36b5:9852:db1d:6ce:d609 prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:13:ae:22 txqueuelen 1000 (Ethernet)
    RX packets 1401 bytes 132143 (129.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 69 bytes 11866 (11.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

(cys_714023149064@kali)-[~/linux-practice]

\$ ip addr

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 08:00:27:13:ae:22 brd ff:ff:ff:ff:ff:ff
    inet 10.36.179.18/24 brd 10.36.179.255 scope global dynamic noprefixroute eth0
        valid_lft 1013sec preferred_lft 1013sec
    inet6 2401:4900:67c3:36b5:9852:db1d:6ce:d609/64 scope global temporary dynamic
        valid_lft 7081sec preferred_lft 7081sec
    inet6 2401:4900:67c3:36b5:a00:27ff:fe13:ae22/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 7081sec preferred_lft 7081sec
    inet6 fe80::a00:27ff:fe13:ae22/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

(cys_714023149064@kali)-[~/linux-practice]

\$ ping google.com

```
PING google.com (2404:6800:4007:836::200e) 56 data bytes
64 bytes from lcmaaa-az-in-x0e.1e100.net (2404:6800:4007:836::200e): icmp_seq=1 ttl=116 time=0.124 ms
64 bytes from lcmaaa-az-in-x0e.1e100.net (2404:6800:4007:836::200e): icmp_seq=2 ttl=116 time=0.124 ms
64 bytes from lcmaaa-az-in-x0e.1e100.net (2404:6800:4007:836::200e): icmp_seq=3 ttl=116 time=0.124 ms
64 bytes from lcmaaa-az-in-x0e.1e100.net (2404:6800:4007:836::200e): icmp_seq=4 ttl=116 time=0.124 ms
64 bytes from lcmaaa-az-in-x0e.1e100.net (2404:6800:4007:836::200e): icmp_seq=5 ttl=116 time=0.124 ms
64 bytes from lcmaaa-az-in-x0e.1e100.net (2404:6800:4007:836::200e): icmp_seq=6 ttl=116 time=0.124 ms
64 bytes from lcmaaa-az-in-x0e.1e100.net (2404:6800:4007:836::200e): icmp_seq=7 ttl=116 time=0.124 ms
64 bytes from lcmaaa-az-in-x0e.1e100.net (2404:6800:4007:836::200e): icmp_seq=8 ttl=116 time=0.124 ms
```

CHAPTER 5:

```
64 bytes from lcmaaa-az-in-x0e.1e100.net (2404:6800:4007:836::200e): icmp_seq=5 ttl=116
64 bytes from lcmaaa-az-in-x0e.1e100.net (2404:6800:4007:836::200e): icmp_seq=6 ttl=116
64 bytes from lcmaaa-az-in-x0e.1e100.net (2404:6800:4007:836::200e): icmp_seq=7 ttl=116
64 bytes from lcmaaa-az-in-x0e.1e100.net (2404:6800:4007:836::200e): icmp_seq=8 ttl=116
^C
— google.com ping statistics —
8 packets transmitted, 8 received, 0% packet loss, time 7454ms
rtt min/avg/max/mdev = 56.273/125.707/268.066/68.775 ms

(cys_714023149064@kali)-[~/linux-practice]
$ netstat -tulnp
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program
(cys_714023149064@kali)-[~/linux-practice]
$ ss -tulnp
Netid      State      Recv-Q      Send-Q      Local Address:Port      Peer Address
(cys_714023149064@kali)-[~/linux-practice]
$ traceroute google.com
traceroute to google.com (2404:6800:4007:836::200e), 30 hops max, 80 byte packets
 1  2401:4900:67c3:36b5::76 (2401:4900:67c3:36b5::76)  3.998 ms  5.585 ms  5.274 ms
 2  * * *
 3  2401:4900:0:1527::1 (2401:4900:0:1527::1)  65.979 ms  2401:4900:0:1528::1 (2401:4900:0:1528::1)  65.761 ms
 4  2401:4900:0:6ff::3 (2401:4900:0:6ff::3)  65.563 ms  65.425 ms  65.320 ms
 5  2401:4900:0:6f9::1 (2401:4900:0:6f9::1)  65.226 ms  64.994 ms  71.862 ms
 6  * * *
 7  2404:a800:3a00:1::601 (2404:a800:3a00:1::601)  77.653 ms  77.290 ms  2404:a800:3a00:1::601  76.681 ms
 8  2404:a800::92 (2404:a800::92)  75.826 ms  75.710 ms  75.001 ms
 9  2404:6800:8202:480::1 (2404:6800:8202:480::1)  74.523 ms  2404:6800:8201::1 (2404:6800:8201::1)  80.994 ms
10  2404:6800:8201:280::1 (2404:6800:8201:280::1)  49.589 ms
11  2001:4860:0:1::559c (2001:4860:0:1::559c)  45.907 ms  2001:4860:0:1::564c (2001:4860:0:1::564c)  45.907 ms

(cys_714023149064@kali)-[~/linux-practice]
$ wget https://example.com
--2026-05-14 13:02:23-- https://example.com/
Resolving example.com (example.com)... 2606:4700:83b5:72db:f23b:623:ef6b:ff98, 104.20.23.154
Connecting to example.com (example.com)|2606:4700:83b5:72db:f23b:623:ef6b:ff98|:443 ... connected
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html'

index.html           [  =>  ]           528
2026-05-14 13:02:33 (18.2 MB/s) - 'index.html' saved [528]

(cys_714023149064@kali)-[~/linux-practice]
$
```

```

cys_714023149064@kali: ~/linux-practice
File Actions Edit View Help
Linux
(cys_714023149064@kali)-[~/linux-practice]
$ nslookup google.com
Server:      10.36.179.21
Address:     10.36.179.21#53
Non-authoritative answer:
Name:   google.com
Address: 142.251.43.78
Name:   google.com
Address: 2404:6800:4007:836::200e

(cys_714023149064@kali)-[~/linux-practice]
$ dig google.com

; <<>> DiG 9.20.2-1-Debian <<>> google.com
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 16358
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                28      IN      A      142.251.43.78

;; Query time: 16 msec
;; SERVER: 10.36.179.21#53(10.36.179.21) (UDP)
;; WHEN: Thu May 14 13:03:10 EDT 2026
;; MSG SIZE rcvd: 44

(cys_714023149064@kali)-[~/linux-practice]
$ █
```

```
cys_714023149064@kali: ~/linux-practice
File Actions Edit View Help
(cys_714023149064@kali)-[~/linux-practice]
$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root          1   0.0  0.3 22580 13576 ?        Ss   12:17   0:01 /sbin/init splash
root          2   0.0  0.0      0     0 ?        S    12:17   0:00 [kthreadd]
root          3   0.0  0.0      0     0 ?        S    12:17   0:00 [pool_workqueue_rele
root          4   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-rcu_gp]
root          5   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-sync_wq]
root          6   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-slub_flushwq]
root          7   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-netns]
root         12   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-mm_percpu_wq]
root         13   0.0  0.0      0     0 ?        I    12:17   0:00 [rcu_tasks_kthread]
root         14   0.0  0.0      0     0 ?        I    12:17   0:00 [rcu_tasks_rude_kthread]
root         15   0.0  0.0      0     0 ?        I    12:17   0:00 [rcu_tasks_trace_kthead
root         16   0.0  0.0      0     0 ?        S    12:17   0:00 [ksoftirqd/0]
root         17   0.0  0.0      0     0 ?        R    12:17   0:01 [rcu_preempt]
root         18   0.0  0.0      0     0 ?        S    12:17   0:00 [rcu_exp_par_gp_kthread]
root         19   0.0  0.0      0     0 ?        S    12:17   0:00 [rcu_exp_gp_kthread_wor
root         20   0.0  0.0      0     0 ?        S    12:17   0:00 [migration/0]
root         21   0.0  0.0      0     0 ?        S    12:17   0:00 [idle_inject/0]
root         22   0.0  0.0      0     0 ?        S    12:17   0:00 [cpuhp/0]
root         23   0.0  0.0      0     0 ?        S    12:17   0:00 [cpuhp/1]
root         24   0.0  0.0      0     0 ?        S    12:17   0:00 [idle_inject/1]
root         25   0.0  0.0      0     0 ?        S    12:17   0:00 [migration/1]
root         26   0.0  0.0      0     0 ?        S    12:17   0:00 [ksoftirqd/1]
root         28   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/1:0H-events_hig
root         30   0.0  0.0      0     0 ?        I    12:17   0:00 [kworker/u10:0-events_u
root         33   0.0  0.0      0     0 ?        S    12:17   0:00 [kdevtmpfs]
root         34   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-inet_frag_wq]
root         35   0.0  0.0      0     0 ?        I    12:17   0:00 [kworker/u9:1-flush-8:0]
root         36   0.0  0.0      0     0 ?        S    12:17   0:00 [kauditd]
root         37   0.0  0.0      0     0 ?        S    12:17   0:00 [khungtaskd]
root         38   0.0  0.0      0     0 ?        S    12:17   0:00 [oom_reaper]
root         40   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-writeback]
root         41   0.0  0.0      0     0 ?        S    12:17   0:00 [kcompactd0]
root         42   0.0  0.0      0     0 ?        SN   12:17   0:00 [ksmd]
root         43   0.0  0.0      0     0 ?        SN   12:17   0:00 [khugepaged]
root         44   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-kintegrityd]
root         45   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-kblockd]
root         46   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-blkcg_punt_b
root         47   0.0  0.0      0     0 ?        S    12:17   0:00 [irq/9-acpi]
root         49   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-tpm_dev_wq]
root         50   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-edac-poller]
root         51   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-devfreq_wq]
root         53   0.0  0.0      0     0 ?        S    12:17   0:00 [kswapd0]
root         61   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-kthrotld]
root         65   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-acpi_thermal
root         66   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-mld]
root         67   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-ipv6_addrconf
root         68   0.0  0.0      0     0 ?        I    12:17   0:00 [kworker/u8:1-ipv6_addr
root         73   0.0  0.0      0     0 ?        I<   12:17   0:00 [kworker/R-kstrp]
```

CHAPTER 6:

```
(cys_714023149064@kali)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G   0    1.9G   0% /dev
tmpfs           393M  980K  392M   1% /run
/dev/sda1       19G   15G   3.3G  82% /
tmpfs           2.0G   4.0K  2.0G   1% /dev/shm
tmpfs           5.0M   0    5.0M   0% /run/lock
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-udev-load-credentials.servi
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup-dev-early.se
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-sysctl.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup-dev.service
tmpfs           2.0G   8.0K  2.0G   1% /tmp
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup.service
C_DRIVE        276G  192G   84G   70% /mnt
tmpfs           1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           393M  112K  393M   1% /run/user/1000

(cys_714023149064@kali)-[~]
$ du -sh ~/linux-practice
100K    /home/cys_714023149064/linux-practice

(cys_714023149064@kali)-[~]
$ sudo fdisk -l
[sudo] password for cys_714023149064:
Disk /dev/sda: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x0f525890

Device     Boot      Start          End      Sectors   Size Id Type
/dev/sda1  *                2048    39684095   39682048   18.9G 83 Linux
/dev/sda2                39686142  41940991   2254850    1.1G  f W95 Ext'd (LBA)
/dev/sda5                39686144  41940991   2254848    1.1G 82 Linux swap / Solaris

(cys_714023149064@kali)-[~]
$ dd if=/dev/zero of=disk.img bs=1M count=50
50+0 records in
50+0 records out
52428800 bytes (52 MB, 50 MiB) copied, 0.0419296 s, 1.3 GB/s

(cys_714023149064@kali)-[~]
$ mkfs.ext4 disk.img
mke2fs 1.47.1 (20-May-2024)
Discarding device blocks: done
Creating filesystem with 51200 1k blocks and 12824 inodes
Filesystem UUID: 1f8d49b0-3cbd-4d55-895b-78d34311c722
Superblock backups stored on blocks:
    8193, 24577, 40961
```

```

(cys_714023149064@kali)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G   0    1.9G   0% /dev
tmpfs           393M  980K  392M   1% /run
/dev/sda1       19G   15G   3.3G  82% /
tmpfs           2.0G   4.0K  2.0G   1% /dev/shm
tmpfs           5.0M   0    5.0M   0% /run/lock
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-udev-load-credentials.servi
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup-dev-early.se
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-sysctl.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup-dev.service
tmpfs           2.0G   8.0K  2.0G   1% /tmp
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup.service
C_DRIVE        276G  192G   84G   70% /mnt
tmpfs           1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           393M  112K  393M   1% /run/user/1000

(cys_714023149064@kali)-[~]
$ du -sh ~/linux-practice
100K    /home/cys_714023149064/linux-practice

(cys_714023149064@kali)-[~]
$ sudo fdisk -l
[sudo] password for cys_714023149064:
Disk /dev/sda: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x0f525890

Device      Boot      Start      End  Sectors  Size Id Type
/dev/sda1   *          2048 39684095 39682048  18.9G 83 Linux
/dev/sda2             39686142 41940991 2254850    1.1G  f W95 Ext'd (LBA)
/dev/sda5             39686144 41940991 2254848    1.1G 82 Linux swap / Solaris

(cys_714023149064@kali)-[~]
$ dd if=/dev/zero of=disk.img bs=1M count=50
50+0 records in
50+0 records out
52428800 bytes (52 MB, 50 MiB) copied, 0.0419296 s, 1.3 GB/s

(cys_714023149064@kali)-[~]
$ mkfs.ext4 disk.img
mke2fs 1.47.1 (20-May-2024)
Discarding device blocks: done
Creating filesystem with 51200 1k blocks and 12824 inodes
Filesystem UUID: 1f8d49b0-3cbd-4d55-895b-78d34311c722
Superblock backups stored on blocks:
    8193, 24577, 40961

```

CHAPTER 7:

```
(cys_714023149064@kali)-[~]
$ mkdir mountdir

(cys_714023149064@kali)-[~]
$ sudo mount -o loop disk.img mountdir

(cys_714023149064@kali)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G   0    1.9G   0% /dev
tmpfs           393M 1020K 392M   1% /run
/dev/sda1       19G   15G   3.3G  82% /
tmpfs           2.0G   4.0K 2.0G   1% /dev/shm
tmpfs           5.0M   0    5.0M   0% /run/lock
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-udev-load-credentials.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup-dev-early.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-sysctl.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup-dev.service
tmpfs           2.0G   8.0K 2.0G   1% /tmp
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup.service
C_DRIVE        276G 192G   84G   70% /mnt
tmpfs           1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           393M 112K 393M   1% /run/user/1000
/dev/loop0      42M   46K   39M   1% /home/cys_714023149064/mountdir

(cys_714023149064@kali)-[~]
$ sudo umount mountdir

(cys_714023149064@kali)-[~]
$ sudo blkid
/dev/sda5: UUID="7b6e1583-03db-43f4-a92c-f823ed805a52" TYPE="swap" PARTUUID="0f525890-05"
/dev/sda1: UUID="35960c0a-71fa-49e7-a195-1c1afa84fe0b" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="0f525890-04"

(cys_714023149064@kali)-[~]
$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda   8:0    0  20G  0 disk
├─sda1 8:1    0 18.9G  0 part /
├─sda2 8:2    0    1K  0 part
└─sda5 8:5    0   1.1G  0 part [SWAP]
sr0   11:0   1 1024M  0 rom

(cys_714023149064@kali)-[~]
$ sudo parted
GNU Parted 3.6
Using /dev/sda
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) ^C

(cys_714023149064@kali)-[~]
$ sudo tune2fs -l disk.img
```

```
(cys_714023149064@kali)-[~]
└─$ sudo tune2fs -l disk.img
tune2fs 1.47.1 (20-May-2024)
Filesystem volume name: <none>
Last mounted on: <not available>
Filesystem UUID: 1f8d49b0-3cbd-4d55-895b-78d34311c722
Filesystem magic number: 0xEF53
Filesystem revision #: 1 (dynamic)
Filesystem features: has_journal ext_attr resize_inode dir_index orphan_file filetype e
adata_csum_seed sparse_super large_file huge_file dir_nlink extra_isize metadata_csum
Filesystem flags: signed_directory_hash
Default mount options: user_xattr acl
Filesystem state: clean
Errors behavior: Continue
Filesystem OS type: Linux
Inode count: 12824
Block count: 51200
Reserved block count: 2560
Overhead clusters: 8349
Free blocks: 42805
Free inodes: 12812
First block: 1
Block size: 1024
Fragment size: 1024
Group descriptor size: 64
Reserved GDT blocks: 256
Blocks per group: 8192
Fragments per group: 8192
Inodes per group: 1832
Inode blocks per group: 458
Flex block group size: 16
Filesystem created: Thu May 14 13:07:47 2026
Last mount time: Thu May 14 13:08:32 2026
Last write time: Thu May 14 13:08:48 2026
Mount count: 1
Maximum mount count: -1
Last checked: Thu May 14 13:07:47 2026
Check interval: 0 (<none>)
Lifetime writes: 323 kB
Reserved blocks uid: 0 (user root)
Reserved blocks gid: 0 (group root)
First inode: 11
Inode size: 256
Required extra isize: 32
Desired extra isize: 32
Journal inode: 8
Default directory hash: half_md4
Directory Hash Seed: 51401a29-0131-4fc4-bb8c-ca981ef859a0
Journal backup: inode blocks
Checksum type: crc32c
Checksum: 0x4da4cc04
Checksum seed: 0x94b587bb
Orphan file inode: 12
```

File Actions Edit View Help

(cys_714023149064@kali)-[~]

\$ iostat

Linux 6.11.2-amd64 (kali) 05/14/2026 _x86_64_ (2 CPU)

avg-cpu:	%user	%nice	%system	%iowait	%steal	%idle
	1.44	0.00	2.65	0.15	0.00	95.76

Device	tps	kB_read/s	kB_wrtn/s	kB_dscd/s	kB_read	kB_wrtn
loop0	0.03	0.12	0.00	0.00	386	9
sda	6.25	223.10	27.41	0.00	727074	89332

(cys_714023149064@kali)-[~]

\$ vmstat

procs		memory				swap--		io		-system--		cpu					
r	b	swpd	free	buff	cache	si	so	bi	bo	in	cs	us	sy	id	wa	st	gu
1	0	0	2861128	47356	451736	0	0	222	27	496	2	1	3	96	0	0	0

(cys_714023149064@kali)-[~]

\$ sar

Cannot open /var/log/sysstat/sa14: No such file or directory
Please check if data collecting is enabled

(cys_714023149064@kali)-[~]

\$ free -h

	total	used	free	shared	buff/cache	available
Mem:	3.8Gi	881Mi	2.7Gi	30Mi	487Mi	3.0Gi
Swap:	1.1Gi	0B	1.1Gi			

(cys_714023149064@kali)-[~]

\$ mpstat

Linux 6.11.2-amd64 (kali) 05/14/2026 _x86_64_ (2 CPU)

01:11:48 PM	CPU	%usr	%nice	%sys	%iowait	%irq	%soft	%steal	%guest	%gnice
01:11:48 PM	all	1.44	0.00	2.16	0.15	0.00	0.49	0.00	0.00	0.00

(cys_714023149064@kali)-[~]

\$ uptime

13:11:55 up 54 min, 2 users, load average: 0.02, 0.09, 0.15

(cys_714023149064@kali)-[~]

\$ dstat

Command 'dstat' not found, but can be installed with:
sudo apt install pcp
Do you want to install it? (N/y)N

(cys_714023149064@kali)-[~]

\$ glances

Command 'glances' not found, but can be installed with:
sudo apt install glances

CHAPTER 8:

```
File Actions Edit View Help

(cys_714023149064@kali)-[~]
$ sudo useradd testuser
useradd: user 'testuser' already exists

(cys_714023149064@kali)-[~]
$ sudo passwd testuser
New password:
Retype new password:
No password has been supplied.
New password:
Retype new password:
passwd: password updated successfully

(cys_714023149064@kali)-[~]
$ sudo usermod -m -d /home/testuser testuser
usermod: no changes

(cys_714023149064@kali)-[~]
$ sudo chage -l testuser
Last password change                : May 14, 2026
Password expires                    : never
Password inactive                    : never
Account expires                     : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7

(cys_714023149064@kali)-[~]
$ groups
cys_714023149064 adm dialout cdrom floppy sudo audio dip video plugdev users netdev bluetooth
nner vboxsf kaboxer

(cys_714023149064@kali)-[~]
$ sudo groupadd developers

(cys_714023149064@kali)-[~]
$ newgrp developers
Password:
newgrp: permission denied

(cys_714023149064@kali)-[~]
$ sudo usermod -aG developers testuser

(cys_714023149064@kali)-[~]
$ su testuser
Password:
$
$ ^C
$ exit

(cys_714023149064@kali)-[~]
$ sudo groupdel developers
```

CHAPTER 9:

```
File Actions Edit View Help

(cys_714023149064@kali)-[~]
$ sudo userdel testuser

(cys_714023149064@kali)-[~]
$ cd ~/linux-practice

(cys_714023149064@kali)-[~/linux-practice]
$ echo "Linux Compression Practice" > compress.txt

(cys_714023149064@kali)-[~/linux-practice]
$ bzip2 compress.txt

(cys_714023149064@kali)-[~/linux-practice]
$ bunzip2 compress.txt.bz2

(cys_714023149064@kali)-[~/linux-practice]
$ xz compress.txt

(cys_714023149064@kali)-[~/linux-practice]
$ unxz compress.txt.xz

(cys_714023149064@kali)-[~/linux-practice]
$ tar -cvf archive.tar compress.txt
compress.txt

(cys_714023149064@kali)-[~/linux-practice]
$ zip files.zip compress.txt
adding: compress.txt (stored 0%)

(cys_714023149064@kali)-[~/linux-practice]
$ unzip files.zip
Archive:  files.zip
replace compress.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: ^C

(cys_714023149064@kali)-[~/linux-practice]
$ lzma compress.txt

(cys_714023149064@kali)-[~/linux-practice]
```

CHAPTER 10:

```
File Actions Edit View Help
$ pwsh
PowerShell 7.2.6
Copyright (c) Microsoft Corporation.

https://aka.ms/powershell
Type 'help' to get help.

[cys_714023149064@kali]~/home/cys_714023149064
PS> uname
Linux

[cys_714023149064@kali]~/home/cys_714023149064
PS> hostname
kali

[cys_714023149064@kali]~/home/cys_714023149064
PS> uptime
03:05:28 up 2 min, 2 users, load average: 0.64, 0.59, 0.25

[cys_714023149064@kali]~/home/cys_714023149064
PS> dmesg
[ 0.000000] Linux version 6.11.2-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14.2.0-6) 14.2.0, GNU ld (GNU Binutils for Debian) 2.43.1) #1 SMP PREEMPT_DYNAMIC Kali 6.11.2-1kali1 (2024-10-15)
[ 0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-6.11.2-amd64 root=UUID=35968c0a-71fa-49e7-a195-1c1afa84fe0b ro quiet splash
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
[ 0.000000] BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000001000000-0x000000000dffff] usable
[ 0.000000] BIOS-e820: [mem 0x000000000dffff0000-0x000000000dffffffffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x00000000fec00000-0x00000000fec0ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000fee00000-0x00000000fee0ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000100000000-0x000000011fffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] SMBIOS 2.5 present.
[ 0.000000] DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
[ 0.000000] DMI: Memory slots populated: 0/0
[ 0.000000] Hypervisor detected: KVM
[ 0.000000] kvm-clock: Using msrs 4b564d01 and 4b564d00
[ 0.000005] kvm-clock: using sched offset of 482516985414 cycles
[ 0.000010] clocksource: kvm-clock: mask: 0xffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 881590591483 ns
[ 0.000017] tsc: Detected 2611.174 MHz processor
[ 0.004741] e820: update [mem 0x00000000-0x00000fff] usable => reserved
[ 0.004754] e820: remove [mem 0x000a0000-0x000fffff] usable
[ 0.004767] last_pfn = 0x120000 max_arch_pfn = 0x400000000
[ 0.004813] MTRR map: 3 entries (3 fixed + 0 variable; max 19), built from 8 variable MTRRs
[ 0.004820] x86/PAT: Configuration [0-7]: WB WC UC- WB WP UC- WT
[ 0.004890] CPU MTRRs all blank - virtualized system.
[ 0.004895] last_pfn = 0xe0000 max_arch_pfn = 0x400000000
```